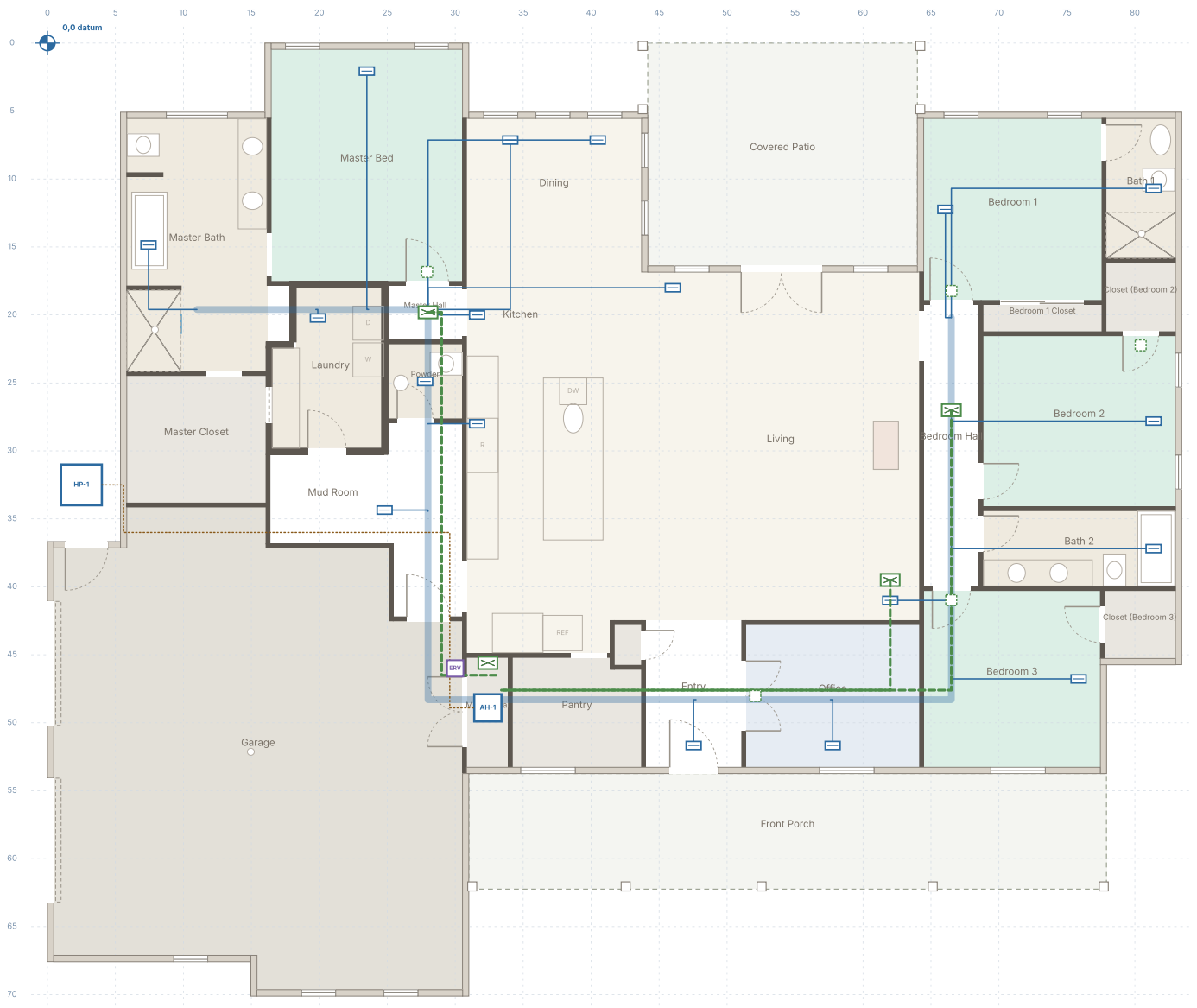
REAR · PATIO



FRONT · PORCH

Supplies blue on the trunks; returns green dashed. · AH-1 in the mechanical closet, HP-1 outside, ERV-1 for fresh air. · 4 tons · heat 40 kBtu/h · cool 21 kBtu/h

draft

M-1 HVAC · 64 Stone Sheep Circle

Room loads at Powell, WY design conditions, one ducted heat pump with backup heat, ducts in the attic. App version v5. From the owner's plans (Larson House, 6/11/26).

Notes

1. Design conditions: Powell, WY (ASHRAE 2009, Cody Muni AWOS (nearest station to Powell), 99.6 % heating / 1 % cooling). Heating -11.4°F, cooling 87.8°F; inside 70°F / 75°F. Location to confirm: these change with it.
2. Envelope: walls U-0.045 (R-20 + R-5 ci), ceiling R-49 (vault R-38), windows U-0.3 SHGC 0.4, slab edge R-10, 3.0 ACH50 (IECC 2021 zone 6).
3. One ducted cold-climate heat pump; the air handler fits the 3'x8' mechanical closet (about 22"x22" footprint, return plenum below or beside). Trunks run in the attic over the flat ceilings: east along the front band and up the bedroom hall, west over the mud room and master hall.
4. The kitchen and living vault (4:12) has no flat attic over it: its supplies are high-sidewall grilles fed from the flat-ceiling attics beside it, or soffit runs.
5. Ducts in the attic: sealed and tested (≤ 4 CFM25 per 100 sf), R-8 insulation, buried under the attic insulation where possible.
6. Every bedroom gets a return path (transfer grille or jump duct). Bath fans and the range hood exhaust outside (E-1); a range hood over 400 CFM needs makeup air.
7. The wood fireplace insert needs its own outside combustion-air kit and a listed chimney.

Open items

- Location: Powell, WY, using design temperatures from Cody (the nearest ASHRAE station, ~700 ft higher, so slightly conservative). Confirm the lot's orientation.
- Fuel: heat pump with electric backup is assumed. A gas furnace with AC is the other common choice; it needs a gas line, a flue and combustion air in the closet.
- Mechanical closet opens from the garage: confirm the AHJ allows the air handler there, and air-seal it (gasketed door, sealed plenum, no return openings).
- Insulation values and air-tightness target for the house.
- Outdoor unit location (drawn by the master closet's outside wall, out of sight from the front).

Room loads and airflow

Room	Floor sf	Ext. wall sf	Glass sf	Heat Btu/h	Cool Btu/h	CFM	Registers	Duct
Master Bed	240	102	25	2,785	1,725	113	1	8"
Master Bath	192	247	14	3,455	960	116	1	8"
Master Closet	96	84	0	1,363	271	46	0	—
Master Hall	22	0	0	108	34	4	0	—
Laundry	87	56	0	904	587	39	1	6"

Room	Floor sf	Ext. wall sf	Glass sf	Heat Btu/h	Cool Btu/h	CFM	Registers	Duct
Powder	30	0	0	150	47	5	1	6"
Mud Room	133	0	0	1,236	356	42	1	6"
Dining	136	149	62	3,771	3,031	199	2	7"
Kitchen	370	0	0	2,280	2,370	156	2	6"
Pantry	77	70	16	1,441	694	48	0	—
Mechanical	25	28	0	493	99	17	0	—
Living	524	111	25	5,927	3,095	203	2	7"
Entry	76	40	0	1,223	256	41	1	6"
Coat Closet	6	0	0	30	9	1	0	—
Office	134	99	16	1,977	1,571	103	1	7"
Bedroom Hall	86	0	0	422	133	14	0	—
Bedroom 1	176	214	25	3,424	1,729	115	1	8"
Bedroom 1 Closet	18	0	0	86	27	3	0	—
Bath 1	53	139	0	1,443	211	48	1	6"
Closet (Bedroom 2)	26	46	0	521	84	18	0	—
Bedroom 2	178	88	25	2,363	1,615	106	1	7"
Bath 2	79	50	0	817	170	27	1	6"
Bedroom 3	169	218	16	3,163	1,436	106	1	7"
Closet (Bedroom 3)	26	92	0	916	127	31	0	—

System

Item	Size	Basis
Heating load	40,299 Btu/h	70°F inside, -11.4°F outside
Cooling load	20,637 sensible + 1,651 latent Btu/h	75°F inside, 87.8°F outside
Heat pump	4 tons, 1,600 CFM	sized to carry the heating load at design on the heat pump alone; cold-climate model
Backup heat	5 kW	covers what the heat pump can't at the heating design temperature
Supply registers	18	4×10 to 6×12 ceiling, 6-8" R-8 flex
Ventilation	133 CFM ERV	ASHRAE 62.2 continuous

Bill of materials

Rough-order quantities from the layout, for pricing. The HVAC contractor takes off the final list.

Item	Qty	Notes
Heat pump + air handler	1 set	4 tons, cold-climate, variable speed, 5 kW strips
Supply registers	18	ceiling and high sidewall, with boots
Return grilles	3	filter grille in the hall, living and master hall
Transfer grilles / jump ducts	5	bedrooms and office
Trunk duct	114 ft	rigid sheet metal, sealed, R-8 wrap
Flex duct, 6-8"	231 ft	R-8, + 25 %
Return duct	~60 ft	14-16" R-8
ERV	1	133 CFM
Line set + pad + disconnect	1	~40 ft